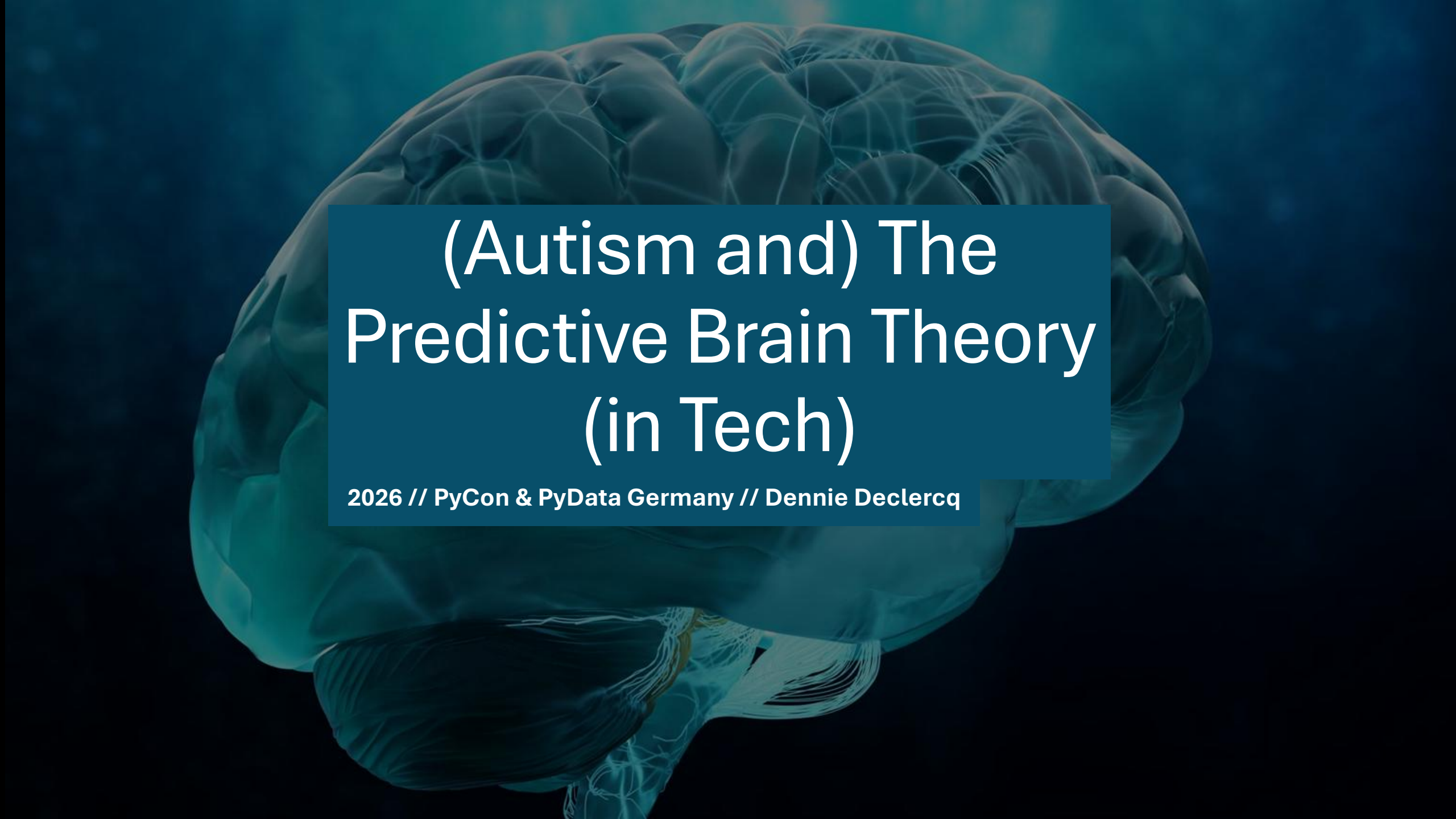




# (Autism and) The Predictive Brain Theory (in Tech)

2026 // PyCon & PyData Germany // Dennie Declercq



# I am...

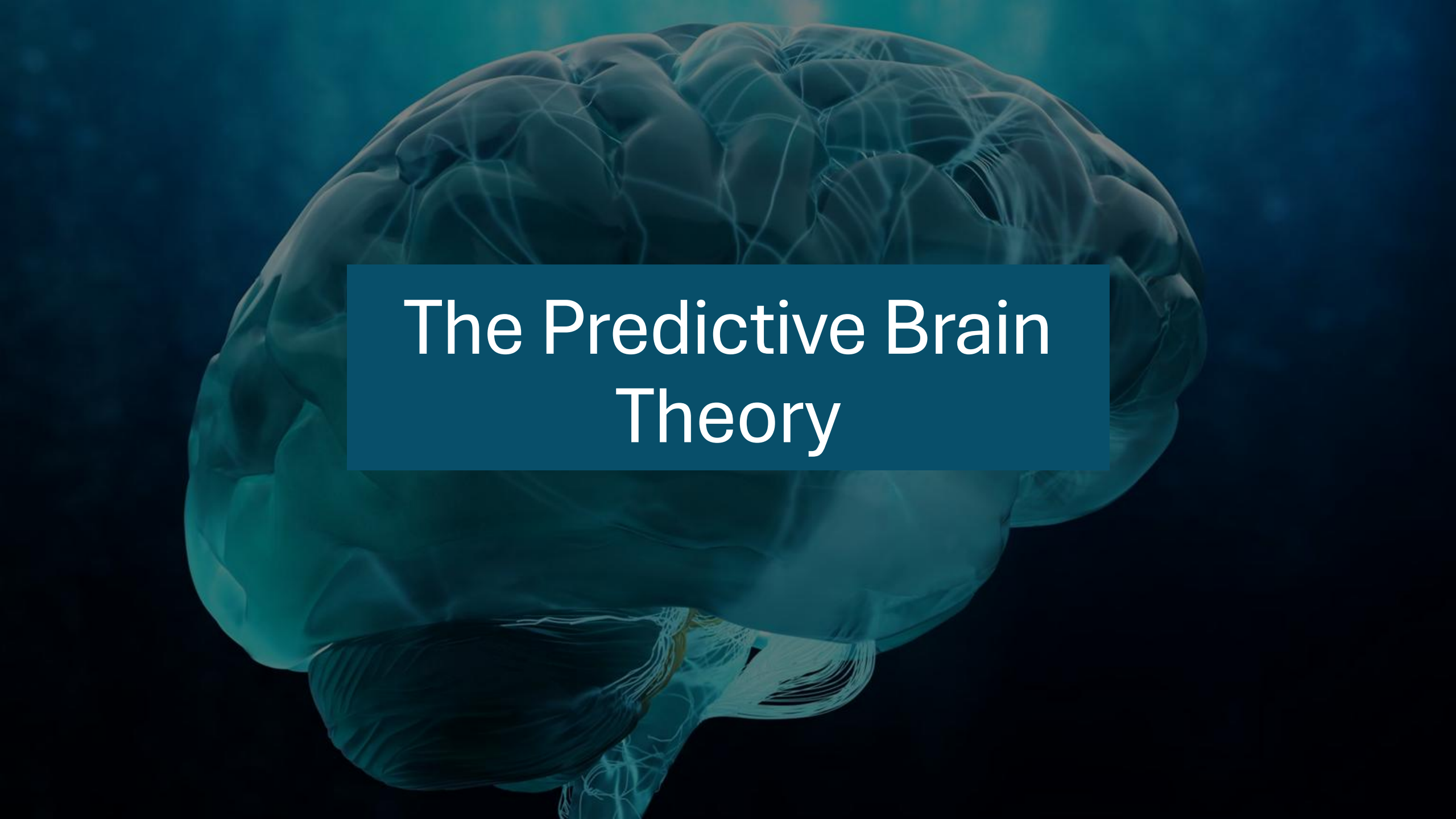
- Dennie Declercq
- President & IT-Coach DDSOFT
- Person with Autism
- Mr. #DreamingIsBelieving
- Microsoft MVP



# Content

- The Predictive Brain Theory
- Autism at the workplace explained by The Predictive Brain Theory





# The Predictive Brain Theory



# Our old believes coded in Computers

Input

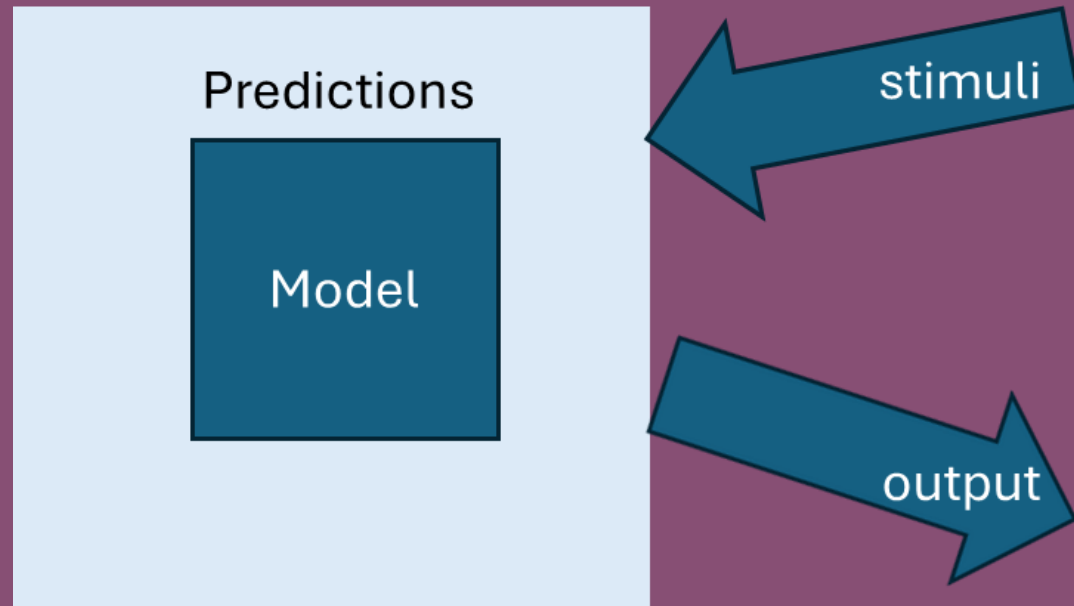
Process

Output

# The Biggest Lie Ever

- During human life there is no clean start.
- Nothing begins “from zero”.
- **Input** (“tell me..”), **Process** (thinking...) does not come from scratch.
- Although: The **input** → **process** → **output** assumes nothing before input.

# A new way of thinking & AI



## AI System

- Model
  - All the names in a region
- Stimuli
  - Execute the question: How to name my child
- Prediction
  - Prediction of relevant names
- Output
  - “Ben”

## Human

- Model
  - All past experiences & learnings
- Stimuli
  - A question, situation or trigger
- Prediction
  - What will be the result
- Output
  - behavior

# A new way | Model

- Our model is the “view of the world”
- Model of the world
- What we learned
- What we experienced
- All the results of our behavior
- The good, the bad and the (very) ugly



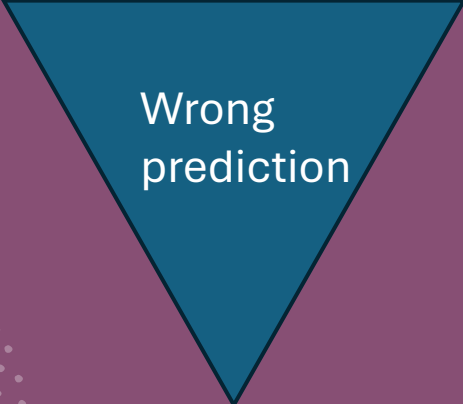
# A new way | Stimuli

- Someone says or behaves to us
- News or notifications
- Announcements
- Changes of rules
- Questions
- Tasks we get

# A new way | Prediction



Right Prediction



Wrong  
prediction

- Based on our model we predict
- We constantly predict different options based on stimuli
- We try to predict accurate which action will lead to good outcome
- We are predicting in hope to make outcome as good as possible

# A new way | Output

- Our output is our behavior
- The decisions we make
- What the outside world / The Business sees

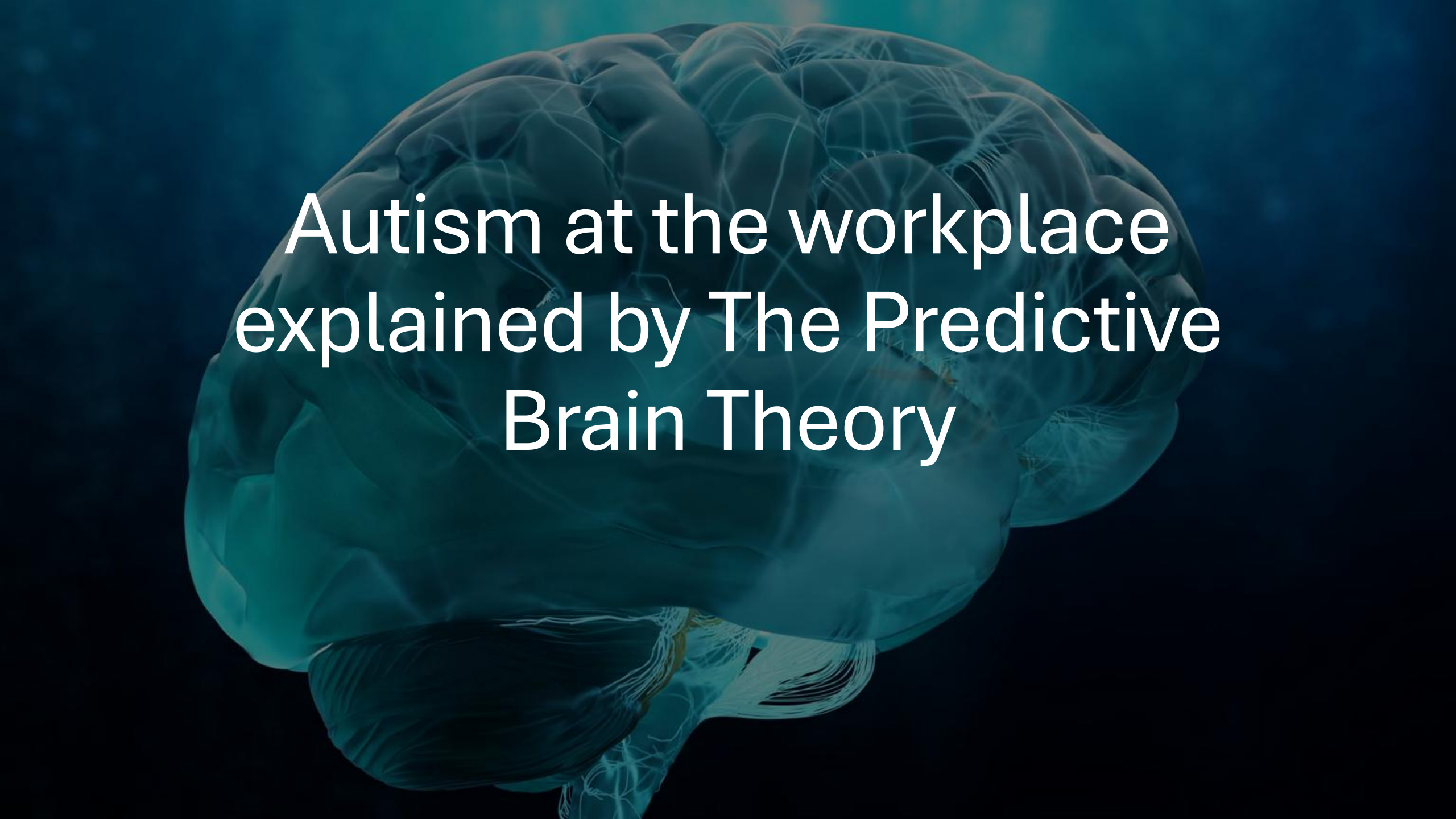
# Executive briefing comparing Old vs New way

- We don't receive input on a blank canvas.
- We don't just process fresh input to have an outcome
- We have a huge model of the world
- On this Model we make predictions
- Inputs are causing our lookups and predictions from the Model
- On those predictions we make our output

# How Autistic Individuals differ from non-autistic individuals

- Recent hypothesis suggest autistic individuals have following differences:
  - A too accurate model of the world causing more prediction errors
  - Limited information about a new situations causes anxiety
  - Behavior based on a prediction error is often seen as 'bad' behavior
  - We need more good prediction results to change our model





# Autism at the workplace explained by The Predictive Brain Theory

# Our Model is trained from past experiences

## Past experiences:

- Bully at the workplace because misunderstood
- Not having the position, we wanted due to bad resume
- Our honesty is being misused
- Software updates gone bad
- Team changes felt brutal

## Has impact at our current job:

- We hide our autism and true selves
- We might not dare to go for promotion
- We don't dare to say if we are feeling overwhelmed
- We unwilling to update our stack
- We block if our leadership changes

# Confirmation & Confirmation Bias

## Our model wants to confirm:

- Last software update crashed the system, this will always be the case
- Last lay-off I was one of the unlucky ones, this will always be the same
- Our customer complained last sprint review, I am afraid for the new review
- Our last off-site team activity gave me a panic attack, I don't want to go again



## So, our Bias causes:

- Being very nervous before the update, lack of sleep and bad concentration causes new crash
- Being unsure and being less productive being in the new lay-off lot
- The customer complains again, this time about my attitude
- The new off-site is mandatory, I am the whole day not OK.

# Stimuli as fuel

When a  
new  
stimuli  
get  
triggered:

An unexpected demand  
A long leave from a colleague  
We get asked for a 1-on-1



We make  
a  
prediction

That can be true  
That can be a prediction-error

# Different kinds of output

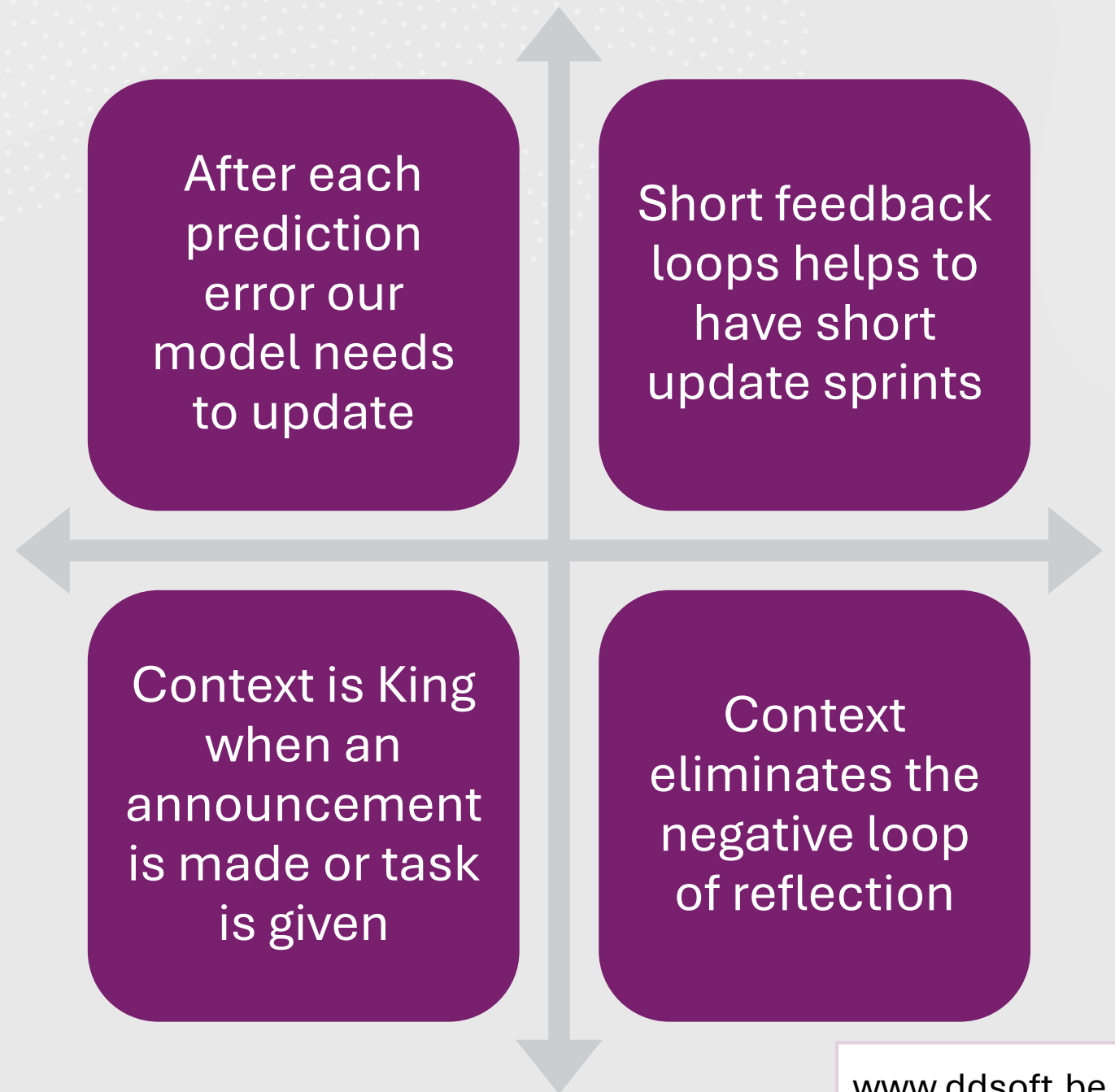
- Predicted good result (positive) + outcome is good (positive)
- Predicted good result (positive) + outcome is bad (negative)
- Predicted bad result (negative) + outcome is bad (negative)
- Predicted bad result (negative) + outcome is good (negative)



**Positive prediction with positive outcome =**  
GOOD FLOW & GROW  
**Negative result with negative outcome =**  
Confirmation to fail mindset and being depressed,  
upcoming potential sickness leave...

**Positive prediction with  
negative outcome =**  
Disappointment, delusion  
**Negative prediction with  
positive outcome =**  
Welcome monkey  
mind

# The Key: Good predictions & Model Finetuning



# Tips to accelerate accurate predictions

When you give a new task → Give context and clear expectations

When a major software (update) bug happens → Do research and postmortem. Communicate

Give planned structured review, don't ask for ad hoc 1-on-1

When another colleague get an exception → Give context, communicate

Be honest on an interview what the job will be and what not

Provide a transparent promotion plan for everyone





# Thank you! || Q&A

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